



Bellkor's Pragmatic Chaos with the winners' check for the Netflix Prize

movies rated based on the age of the movies. The key idea behind doing this is that the time at which a movie is released and the impression and rating it would have gotten immediately after its release is vastly different from the kind of

rating it will receive later. To take an example, a movie like the original *Die Hard* (1988), is more likely to receive higher ratings now than it would have at the time of its release because of its growth as a cult classic. Some movies simply age better than others. By noticing the fact that users use different criteria to rate movies from different time periods, the algorithms took a huge step forwards in improving Cinematch. Another important factor they took into consideration was the time at which the rating was given. While this may seem pedantic and inconsequential, the cumulative effect of a million ratings all done on Monday, when viewers are less happy and are therefore less likely to rate a movie highly, is quite huge. The winning algorithms are a linear blend of two algorithms: Restricted Boltzmann Machines, a form of neural networking and the so called SVD++, a form of matrix factorization which was developed by the winning team to account for implicit information in the data. These small steps combined together pushed the Bellkor algorithm over the 10% mark. It would be unfair to mention the Netflix prize without lauding the efforts of the Runner-up team, 'The Ensemble', who achieved the same score as the Bellkor team, but were late by a mere 24 minutes. When 24 minutes was

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